Data Analytics

Real-Time and Streaming Analytics

When the data size is massive, generally it is being subjected to batch and historical processing techniques to extract all kinds of hidden intelligence. However, on the other hand, there are fast and streaming data which are relatively small in size but widely vary in structure and speed. There are open source and commercial-grade platforms emerging and evolving for undertaking real-time and streaming analytics on fast and streaming data. Recently, there has risen demand for doing immediate and interactive analytics not only on fast data but also on big data to emit out insights in time. With the grand availability of exemplary solutions, systems, and services, the days of accomplishing real-time analytics on big data are not too far away. All these advancements do not imply that the fully matured and stabilized BI and DW solutions are going away. But there is a telling need to complement them with newer technologies for big, fast, streaming, and IoT data storage, management, and analytics. The era of real-time and streaming analytics is steadily arriving, and there are a number of use cases being unearthed mandating real-time and streaming analytics capabilities and competencies for any organization to grow fast. There are well-known use cases such as full-text indexing, recommendation systems (e.g., Netflix movie recommendations), log analysis, computing web indexes, and data mining. These are the well-known processes that can be allowed to run for extended periods of time

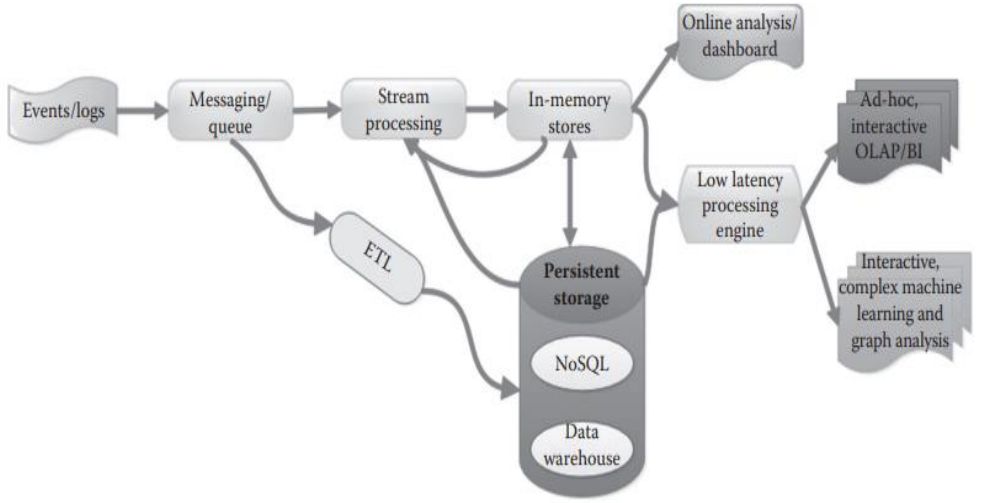

Figure 7.2 The real-time and streaming analytics architecture.

Operational analytics (OA) is a kind of real-time analytics that provides enhanced visibility into business processes, events, and operations as they are happening. The practice of OA is succulently enabled by special technologies that can handle machine data, sensor data, event streams, and other

forms of streaming data and big data. OA solutions can also correlate and analyze data collected from multiple sources in various latencies (from batch to real time) to realize and reveal actionable information. Organizations can act on extracted knowledge by sending an appropriate alert notification to the correct manager in time, updating a management dashboard, offering an incentive to a churning customer, adjusting machinery, or preventing fraud.

Data Analytics Platforms

These are a bit complicated platforms directly attached with AEPs. There are powerful data mining, fusion, and processing algorithms getting implemented and incorporated in these platforms to elegantly draw knowledge out of data heaps. The insights extracted are then fed into AEPs to drive pioneering applications. The challenge with these platforms is that the amount of sensor and actuator data is becoming massive, and hence doing real-time analytics on big data is a bit difficult. There are technologically sound and solid solutions emerging in order to tackle this particular problem. The underlying IT infrastructures also need to be exceptionally elastic enough to cope with the big data challenges. Increasingly data storage, processing, mining, and analyzing software solutions are being deployed in private, public, or even in hybrid clouds. Data are subjected to a variety of deeper and decisive investigations in order to extract pragmatic insights. There are big, fast, streaming, and IoT data whereas there are batch or ad hoc or historical or comprehensive processing apart from interactive and iterative processing. Thus, there are specific as well as integrated platforms on cloud environments for embarking on big and fast data analytics. Especially predictive, prescriptive, and personalized analytics domains are expected to be dominant in the years ahead.

Local Analytics

Local analytics, entails processing some analytical information at the gateway, so that only meaningful information is sent to the next tier which could be another gateway, the data center, or the cloud. This minimizes consumption of expensive network bandwidth and reduces overall solution latency.

Cloud Analytics

Cloud analytics is the use of remote public or private computing resources—known as the cloud—to analyze data on demand. Cloud computing analytics helps streamline the business intelligence process of gathering, integrating, analyzing, and presenting insights to enhance business decision making.

How cloud analytics works: Cloud analytics works by allowing a business to use the advanced data analytics tools available on cloud analytics platforms to analyze vast quantities of data. Businesses can then report and store those findings for repeat use. Cloud analytics offerings are

typically offered as a subscription or pay on a volume of data or query basis. Cloud analytics has proven to be a faster way to gain business-critical insight for decision making.

Advantages of cloud analytics: Advantages of cloud analytics include helping businesses more efficiently process and report data findings, enhance collaboration, and provide decision-makers faster access to business intelligence.

Scalability & agility: Scalability, in volume of users and data, is one of the advantages of cloud analytics. Another is the increased agility to ask new questions on demand. Cloud computing can enable the abilities to add data storage and data analysis capacity as needed. In this way, businesses can most efficiently scale the storing, processing, and leveraging of data through insights that reflect changing market conditions.

Unified approach to data: Cloud analytics can provide a unified approach to data, allowing fast access to necessary business data by your organization's decision-makers. That includes access to different types of data, as well as data from different sources. You can rapidly explore, test, and evaluate data to find strategic insights that advance the enterprise.

Breaks down silos: When a business takes advantage of cloud analytics, it can effectively achieve cross-organizational integration of data. Leveraging data from different parts of your organization through a cloud analytics solution can deliver daily insights. These insights can be a competitive advantage that propels your business forward. Data stored and analyzed in the cloud also makes it easier for employees of all types to quickly access data and share insights, supporting effective communication, collaboration, and decision making.

Find answers and draw insights quicker: A cloud analytics platform allows businesses to better integrate—and analyze—data quickly to identify intelligent insights for actionable decisions at the moment of relevance.

Encourage more collaboration: When businesses use a cloud analytics platform, they gain a central place to access data and connect to shared information whenever necessary or proactively delivered. This shared connection and fast access to data enhance collaboration throughout the enterprise.

Increase data security and governance: A cloud-based analytics platform can provide a single and secure access point to data. Also, a cloud analytics platform improves data governance by giving you more granular access controls for who has access to what data and audit capabilities to understand who accessed what data.